

■ B.4.4 Recursion Limits

Mathematica usually goes on evaluating an expression according to the procedure described above until the expression no longer changes.

There are, however, some expressions for which this procedure will never terminate. For example, if you make the assignment $x=x+1$, then ask for the value of x , the standard evaluation procedure will lead to an infinite loop.

Mathematica has various mechanisms which protect it from this kind of situation.

First, you can always interrupt the operation of *Mathematica*, as discussed in Section ??.

In addition, *Mathematica* has a built-in “recursion limit”, which stops evaluation after a fixed number of “levels”. If *Mathematica* reaches the recursion limit, it returns the expression it is trying to evaluate, wrapped in a `Hold` function to prevent further evaluation.

<code>\$RecursionLimit</code>	the current setting of the recursion limit
<code>\$RecursionLimit=$n$</code>	set the recursion limit to n levels
<code>\$RecursionLimit=Infinity</code>	allow unrestricted recursion

Functions for manipulating the global recursion limit.

The exact counting of “recursion levels” depends on many details of the *Mathematica* definitions you are using. In a typical case, one recursion level is used every time a function that depends on another function is evaluated.